Power series solutions

Now that we understand how to add series, we can start looking at series solutions for differential equations.

When a power series solution is defined

Our goal is to use series to solve second order linear homogeneous differential equations.

$$p(x)y'' + q(x)y' + r(x)y = 0$$

And when we find series solutions, we can handle nonconstant coefficients, so $p(x)$, $q(x)$, and $r(x)$ don't need to be constants. Specifically, we're going to look at series solutions when $p(x)$, $q(x)$, and $r(x)$ are polynomial functions.

As long as $q(x)/p(x)$ and $r(x)/p(x)$ are **analytic**, which means that we can define Taylor series for $q(x)/p(x)$ and $r(x)/p(x)$ around $x = x_0$, then we say that $x = x_0$ is an **ordinary point**. As long as we stick to the case where $p(x)$, $q(x)$, and $r(x)$ are polynomials, then $x = x_0$ is an ordinary point and $q(x)/p(x)$ and $r(x)/p(x)$ are analytic as long as $p(x_0) \neq 0$. When $x = x_0$ is non-ordinary, we call it a **singular point** instead.

We can always find two linearly independent power series solutions around an ordinary point, where both solutions are in the form

$$y(x) = \sum_{n=0}^{\infty} c_n(x - x_0)^n$$

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When $x_0 = 0$, we can simplify these power series solutions to

$$y(x) = \sum_{n=0}^{\infty} c_n x^n$$

The solutions will converge at least on the interval $|x - x_0| < R$, where R is the minimum radius of convergence, and represents the distance from x_0 to the closest singular point.

Keep in mind that this series solution extends to nonhomogeneous differential equations as well,

$$p(x)y'' + q(x)y' + r(x)y = g(x)$$

as long as $g(x)$ is also analytic around $x - x_0$.

Let's do an example so that we understand how to find this minimum radius of convergence for the solutions.

Example

Find the minimum radius of convergence of a power series solution of the differential equation about the ordinary point $x_0 = 1$.

$$(x^2 - 25)y'' + 2xy' + y = 0$$

The singular points of the differential equation occur where $x^2 - 25 = 0$, or at $x = \pm 5$. In the plane, we consider the ordinary point $x_0 = 1$ to occur at $(1,0)$, and the singular points $x = \pm 5$ to occur at $(5,0)$ and $(-5,0)$. The

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distance from the ordinary point $(1,0)$ to the closer singular point $(5,0)$ is 4, and the distance from the ordinary point $(1,0)$ to the further singular point $(-5,0)$ is 6. So the minimum radius of convergence is

$$|x-1|<4$$

Finding a power series solution

In order to find series solutions, we'll need the derivatives of the power series. The power series in x and its first two derivatives are

$$y=\sum_{n=0}^{\infty}c_nx^n=c_0+c_1x+c_2x^2+c_3x^3+\ldots$$

$$y'=\sum_{n=1}^{\infty}c_nnx^{n-1}=c_1+2c_2x+3c_3x^2+4c_4x^3+\ldots$$

$$y''=\sum_{n=2}^{\infty}c_nn(n-1)x^{n-2}=2c_2+6c_3x+12c_4x^2+\ldots$$

To find a power series solution, we'll substitute these series into our differential equation for y, y', and/or y''. Then we'll use what we learned in the last lesson to add power series by first making sure that they're in phase and have matching indices.

Once we add the series, we want to factor the result and then define the **recurrence relation** equation, which is an equation that defines values in the series in terms of previous values in the series. We'll solve the

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recurrence relation for the coefficient with the largest subscript, and then we'll use that equation to build out the first few terms of the series.

Finally, we'll put those first few terms together, and work out the pattern they follow to build a simplified power series representation, which will be the general solution to the differential equation.

Let's do an example to see what this looks like.

Example

Find a power series solution to the differential equation $y' = xy$.

$$y = \sum_{n=0}^{\infty} c_n x^n$$

To start, rewrite the differential equation.

$$y' = xy$$

$$y' - xy = 0$$

Next, we'll substitute y and y' into the differential equation.

$$\sum_{n=1}^{\infty} c_n n x^{n-1} - x \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=1}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

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Let's check to see if the series are in phase. Because the index of the first series starts at $n = 1$, the first series begins with an x^0 term, and because the index of the second series starts at $n = 0$, the second series begins with an x^1 term. So we'll pull the x^0 term out of the first series.

$$c_1 + \sum_{n=2}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

Now both series are in phase, but the indices don't match. We can substitute $k = n - 1$ and $n = k + 1$ into the first series, and $k = n + 1$ and $n = k - 1$ into the second series.

$$c_1 + \sum_{k=1}^{\infty} c_{k+1}(k+1)x^k - \sum_{k=1}^{\infty} c_{k-1}x^k = 0$$

With the series in phase and matching indices, we can finally add them.

$$c_1 + \sum_{k=1}^{\infty} (c_{k+1}(k+1)x^k - c_{k-1}x^k) = 0$$

$$c_1 + \sum_{k=1}^{\infty} (c_{k+1}(k+1) - c_{k-1})x^k = 0$$

The c_1 value in front of the series is associated with the $k = 0$ term, while all of the other $k = 1, 2, 3, \ldots$ terms are still in the series.

$$k = 0 \qquad c_1 = 0$$

$$k = 1, 2, 3, \ldots \qquad c_{k+1}(k+1) - c_{k-1} = 0$$

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The equation $c_{k+1}(k+1) - c_{k-1} = 0$ is the recurrence relation, and we always want to start by solving it for the coefficient with the largest subscript. This recurrence relation includes c_{k+1} and c_{k-1}, and since $k+1 > k-1$, we'll solve the equation for c_{k+1}.

$$c_{k+1} = \frac{c_{k-1}}{k+1}$$

Now we'll start plugging in values $k = 1, 2, 3, \ldots$.

$$k = 1 \quad c_2 = \frac{c_0}{2} \qquad k = 2 \quad c_3 = \frac{c_1}{3}$$

$$k = 3 \quad c_4 = \frac{c_2}{4} = \frac{c_0}{2 \cdot 4} \qquad k = 4 \quad c_5 = \frac{c_3}{5} = \frac{c_1}{3 \cdot 5}$$

$$k = 5 \quad c_6 = \frac{c_4}{6} = \frac{c_0}{2 \cdot 4 \cdot 6} \qquad k = 6 \quad c_7 = \frac{c_5}{7} = \frac{c_1}{3 \cdot 5 \cdot 7}$$

$$\vdots \qquad \vdots$$

$$c_{2k} = \frac{c_0}{(2)(4)(6)\ldots(2k-1)(2k)} \qquad c_{2k+1} = \frac{c_1}{(3)(5)(7)\ldots(2k)(2k+1)}$$

for $k = 1, 2, 3, \ldots$ for $k = 1, 2, 3, \ldots$

But because we know $c_1 = 0$, all the terms on the right side of our table that are defined in terms of c_1 (all of the even values of k) will go to 0.

$$c_1 = 0$$

$$k = 1 \quad c_2 = \frac{c_0}{2} \qquad k = 2 \quad c_3 = 0$$

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$k = 3$ $\quad c_4 = \dfrac{c_2}{4} = \dfrac{c_0}{2 \cdot 4}$ $\qquad$ $k = 4$ $\quad c_5 = 0$

$k = 5$ $\quad c_6 = \dfrac{c_4}{6} = \dfrac{c_0}{2 \cdot 4 \cdot 6}$ $\qquad$ $k = 6$ $\quad c_7 = 0$

$\vdots$ $\qquad$ $\vdots$

$c_{2k} = \dfrac{c_0}{(2)(4)(6)\ldots(2k-2)(2k)}$ $\qquad$ $c_{2k+1} = 0$

for $k = 1, 2, 3, \ldots$ $\qquad$ for $k = 1, 2, 3, \ldots$

Using these values, the solution is

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + c_6 x^6 + c_7 x^7 + \ldots + c_{2k} x^{2k} + c_{2k+1} x^{2k+1}$$

$$y = c_0 + \frac{c_0}{2} x^2 + \frac{c_0}{2 \cdot 4} x^4 + \frac{c_0}{2 \cdot 4 \cdot 6} x^6 + \ldots + \frac{c_0}{(2)(4)(6)\ldots(2k-2)(2k)} x^{2k}$$

Factor out the c_0 coefficient.

$$y = c_0 \left(1 + \frac{1}{2} x^2 + \frac{1}{2 \cdot 4} x^4 + \frac{1}{2 \cdot 4 \cdot 6} x^6 + \ldots + \frac{1}{(2)(4)(6)\ldots(2k-2)(2k)} x^{2k} \right)$$

To figure out how to represent the series, we realize that we have a value of 2 multiplied through the factorial part of the denominator, so we can pull that out, representing it as 2^k.

$$y = c_0 \left(1 + \frac{1}{2^1(1)} x^2 + \frac{1}{2^2(1 \cdot 2)} x^4 + \frac{1}{2^3(1 \cdot 2 \cdot 3)} x^6 \right.$$

$$\left. + \ldots + \frac{1}{2^k(1 \cdot 2 \cdot 3 \cdot \ldots \cdot (k-1) \cdot k)} x^{2k} \right)$$

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Then we'll replace the factorial portion with $k!$, and we know also that every x^2, x^4, x^6, ... term can be represented by x^{2k}, so

$$y = c_0\left(1 + \frac{1}{2^1(1)}x^2 + \frac{1}{2^2(1\cdot 2)}x^4 + \frac{1}{2^3(1\cdot 2\cdot 3)}x^6 + \ldots + \frac{1}{2^k\cdot k!}x^{2k}\right)$$

And the pattern that seems to be emerging is

$$y = c_0\sum_{k=0}^{\infty}\frac{1}{2^k k!}x^{2k}$$

This series representation seems to work for all values $k = 1, 2, 3, \ldots$, but let's make sure it also works for the $k = 0$ term. If we plug $k = 0$ into this series representation, we get

$$\frac{1}{2^0 0!}x^{2(0)} = \frac{1}{(1)(1)}(1) = 1$$

This representation works for every term in the series, so let's just rewrite it in a simpler way to give the general solution to the differential equation.

$$y = c_0\sum_{k=0}^{\infty}\frac{1}{k!}\left(\frac{x^{2k}}{2^k}\right)$$

$$y = c_0\sum_{k=0}^{\infty}\frac{1}{k!}\left(\frac{x^2}{2}\right)^k$$

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Equivalent solution from first principles

The power series solution we found in this last example looks a little complicated, but it's actually equivalent to the solution we would've found if we'd solved the differential equation using separation of variables.

$$\frac{dy}{dx} = xy$$

$$\int \frac{1}{y}\, dy = \int x\, dx$$

$$\ln|y| = \frac{1}{2}x^2 + C$$

$$|y| = e^{\frac{1}{2}x^2 + C}$$

$$y = Ce^{\frac{1}{2}x^2}$$

The Maclaurin series for e^x is

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!}x^n$$

In the solution $y = Ce^{\frac{1}{2}x^2}$, we have $e^{\frac{1}{2}x^2}$ instead of e^x, so let's replace x with $(1/2)x^2$ in the formula for the Maclaurin series representation of e^x.

$$e^{\frac{1}{2}x^2} = \sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{1}{2}x^2\right)^n$$

$$e^{\frac{1}{2}x^2} = \sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{1}{2}\right)^n (x^2)^n$$

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$$e^{\frac{1}{2}x^2} = \sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{1}{2^n}\right)(x^{2n})$$

$$e^{\frac{1}{2}x^2} = \sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{x^{2n}}{2^n}\right)$$

$$e^{\frac{1}{2}x^2} = \sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{x^2}{2}\right)^n$$

If we plug this back into the general solution $y = Ce^{\frac{1}{2}x^2}$, we get

$$y = C\sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{x^2}{2}\right)^n$$

We can see that this series matches exactly the series we found in the last example,

$$y = c_0\sum_{k=0}^{\infty} \frac{1}{k!}\left(\frac{x^2}{2}\right)^k$$

In other words, we're finding the same general solution to the differential equation that we would have found using other methods, we're just now expressing the solution as a series.

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